Market Valuation of Artificial Intelligence Implementation Announcements

Cheng-Kui (Tony) Huang
National Chung Cheng University

Yu-Tsung Lee
National Chung Cheng University

Abstract

With the advent of artificial intelligence (AI), more and more enterprises have begun developing AI to promote their own competitive advantage and increase their business value. In order to reveal the value-added results of AI implementation, this study adopts an event study methodology to capture the abnormal returns resulting from the announcement of AI implementation. Based on the empirical results of this study, in response to the announcement of AI implementation, we find significant positive abnormal returns on the event day. With respect to the grouping analysis, the results show significant differences in average cumulative abnormal returns between announcements with detailed information and those without detailed information. In addition, our findings present significant differences in average cumulative abnormal returns between IT-companies and non-IT companies on the event day. Ultimately, according to the content analysis, only one characteristic, the frequency of negative words, is modestly and negatively correlated with average cumulative abnormal returns.

Keywords: Artificial Intelligence; Business Value of IT; Event Study Methodology; Abnormal Return; Content Analysis.

Introduction

The most important modern general-purpose technology is artificial intelligence (AI), in particular machine learning (Brynjolfsson & Mcafee, 2017). With the rapid advance of AI technology, performance is improving on tasks originally accomplished by human beings. The development and adoption of deep learning techniques have made the learning process quicker and more efficient using improved algorithms and hardware. Accordingly, we expect appropriate utilization of AI techniques by managers to bring huge impacts on business, generating significant benefits for companies.

We refer to the white paper written by SAS, Accenture Applied Intelligence, Intel, and Forbes Insights on the internal impact of AI (SAS et al., 2018), which reveals that of the more than 300 executives from large worldwide companies which have implemented AI, 51 percent say that the deployment of AI-based technologies on their operations has been successful. Moreover, 60 percent of executives express that AI technologies are helpful in accurate forecasting and decision-making; another 25 percent expect to see this benefit in the near future.

We also study trends of AI-related news from Fast and Horvitz (2017) to understand the external impact of AI. An analysis of three million articles in the New York

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Times from 1986 to 2016 shows that AI discussion has exploded since 2009; in addition, pro-AI news has consistently grown two to three times more than that of pessimistic news.

We thus see that AI implementation is influential and helpful for enterprises. Although various reports have praised the power of AI, value-added results for enterprises implementing AI remain unclear. To the best of our knowledge, there are no studies so far able to proffer certain evidence to reveal the influence of AI on enterprises in quantitative terms. Such evidence is necessary for managers to convince their shareholders (investors) and stakeholders to believe that AI investment can create new revenue for their companies.

Based on this argument, in this study, we investigate the business value of AI from a market perspective by adopting an event study methodology. Drawing upon the efficient-market hypothesis (Fama, 1970) and signaling theory (Connelly et al., 2011), the market fully reacts to the available signals and unexpected events in public media, and the reactions are based on the tones of these events. Hence, these reactions lead to the fluctuation of stock prices and abnormal returns. With event study methodology, managers acquire the perceived expectations from shareholders (investors) with respect to their AI implementation and measure the business value of AI investment.

In the previous studies of business value of information technology (IT), most findings showed that the investments of transformational ITs have a positive impact on firms' market value (Subramani and Walden, 2001). Although AI is also one kind of transformational ITs, there is no such information technology like AI, sparking various debates. Some people enjoy the advantages of AI on saving cost and improving performance; however, others doubt AI because it still has numerous risks and challenges (Cao et al., 2021; Future of life, 2021). For example, AI might have hidden bias or discrimination for making right decisions because its rules/patterns may inherit human prejudice, stored in the form of big data (Duan et al., 2019; Shrestha et al., 2019). Additionally, AI brings the issue of potential job losses or technological unemployment so that human beings feel threatened and might oppose AI (Jarrahi, 2018; Ransbotham et al., 2018). Hence, whether the investment of AI is optimistic or pessimistic from the perspective of investors remains a question and is worthy to be investigated further.

Following the news selection principles of Tetlock et al. (2008) and Konchitchki and O'Leary (2011), we select 109 AI implementation announcements from 2014 to 2019 as events for the analysis of abnormal returns. In addition, to understand what kinds of characteristics of AI implementation announcements are related to abnormal returns, we group these events in four classes to analyze their impacts. These classes are (1) whether the announcement offers detailed information, (2) whether the announcement mentions technological partnerships, (3) whether the company is an IT or non-IT company, and (4) whether different types of AI can lead to different outcomes.

Finally, we also evaluate the impact in terms of the content characteristics of the announcement to investigate the impact of AI implementation. According to Tetlock (2007), negative words are more influential than other types of words in the Harvard-IV-4 psychosocial dictionary. Likewise, in reference to the study of Pitler and Nenkova (2008), we know that vocabularies largely determine readability; higher readability makes texts easier to read and helps the audience to clearly receive the signal that texts seek to deliver. Thus, we employ the readability scores from the READABLE.IO web tool to measure the readability of each announced new coverage about AI implementation. The scores are evaluated using the Flesch-Kincaid grade level, the Gunning Fog index, the Coleman-Liau index, the automated readability index, and SMOG (Ghose & Ipeirotis, 2011). After that, we average the five scores of each event and construct a regression model to observe the correlation between cumulative average abnormal returns and content characteristics.

The contribution of this study is to proffer evidence for enterprises to understand whether AI initiatives bring benefits from an investment perspective. In addition, we seek to inform academic scholars that the research direction of AI is not merely creating or improving AI algorithms, but exhibiting greater business values via AI for managers and shareholders.

This paper is organized as follows: Section 2 reviews related work; Section 3 develops our hypotheses; Section 4 introduces the research methodology; Section 5 presents our findings; and finally, Section 6 discusses implications, limitations, future work, and then concludes.

Related Work

In this section, we discuss related work, starting in artificial intelligence and event study methodology. Then, we discuss measures of market reaction, and content analysis.

Artificial Intelligence

Artificial intelligence (AI) is any technique that enables computers to mimic human behavior; thus, AI is relevant to any intellectual task and constitutes a universal field. Currently, AI encompasses a variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, autonomous driving, and disease diagnosis (Russell & Norvig, 2016). Moreover, according to IDC (International Data Corporation), a research firm, the AI-related market is predicted to surge to $47 billion by 2020 (Lohr, 2016).

Although AI is now one of the most popular technologies, it has been under development since the 1950s. In the 1980s, machine learning, an AI technique, was developed to give a computer the ability to learn without explicit programming. Then, in the 2010s, deep learning was proposed, a subset of machine learning which makes feasible the computation of multi-layer neural networks. The results of deep learning are comparable to human experts. One famous example is AlphaGo, which won against an 18-time world Go champion.

To distinguish AI ability, the terms strong AI and weak AI were coined by John Searle (Searle, 1980). Strong AI thinks and has a "mind", whereas weak AI only focuses on doing one special thing. In current AI development, most applications designed to solve problems for human beings are still the latter type. Currently, the former is not easily achieved.

Concerning managerial issues of AI, Tambe et al. (2019) discuss a gap between the promise and reality of AI in human resource (HR) management and identify challenges in utilizing data science technologies for HR tasks. Huang and Rust (2018) propose the theory of AI job replacement, as AI reshapes service; accordingly, a new type of service with AI is to be recognized. In addition, to measure the economic value of AI (machine learning), Brynjolfsson et al. (2019) discuss how machine translation can affect international trade on eBay, a digital platform.

Recent literature shows that discussion of AI is diverse and not limited to technological issues. In addition, due to the increasing number of applications and achievements of AI and optimistic AI-related news coverage (Fast & Horvitz, 2017), investors have become more familiar with AI technology and have begun to focus on firms' AI implementation announcements. Hence, in this study, we adopt the event study method to understand the business value of AI.

Applications of Market Value Study

According to Subramani and Walden (2001), scholars previously used traditional productivity measurement approaches and financial indicators to measure the value of investments in information technology (IT) or information systems (IS). Their results indicate that scholars believe that the impact of IT/IS is indirectly connected to financial performance, and that IT/IS concerns efficiency only and not effectiveness. However, due to the rapid advancement and immeasurable influence of IT/IS, such approaches are no longer appropriate.

In accordance with McWilliams and Siegel (1997), event study methodology is a powerful tool that helps researchers to assess the changing financial impact of corporate policy based on the stock market, an outside perspective. As the basis for event study methodology, the efficient-market hypothesis (EMH) is a theory in financial economics stating that stock prices fully reflect all available information (Fama, 1970). That is, when unexpected news about a particular company is announced on public media and reaches the capital market, investors evaluate the relevance and influence of the news with respect to the company (Malkiel & Fama, 1970). Thus, the stock price is the value of the company, including that value perceived by investors. Furthermore, the study of Son et al. (2014) shows that the event study method has evolved to measure the market reaction to initiative actions of companies and capture the differences of market values of companies when specific announcements are released.

To the best of our knowledge, Dos Santos et al. (1993) published the first study using event study methodology to evaluate the business value of IT/IS in the management information systems (MIS) field. Since then, MIS researchers have begun to increasingly adopt this methodology to investigate the impact of different types of IT/IS (Konchitchki & O'Leary, 2011; Roztocki & Weistroffer, 2011). Table 1 shows recent representative literature that applies the event study method.

In previous studies using event study methodology for IT/IS investments, as shown above there are various types of IT/IS. However, to the best of our knowledge, no studies investigate the influence of AI initiatives on the abnormal return. This is however an imperative and popular issue for practitioners and researchers, leading us to address the research gap. In addition, these ITs/ISs can be classified into two groups according to their functions: operation and strategy (Huang et al., 2016). In this study, we use event study methodology to analyze announcements of AI implementation, which helps managers to relocate their company position in the market and formulate new strategies. Accordingly, we argue that investment in AI is a strategic function, which can benefit companies in strategic management if these companies aim to obtain more competitive advantage in their industry.

Table 1. Recent Literature Using Event Study Method

Topic	Description	Source
Identity theft countermeasures (ITC)	The authors analyze the short- and long-term impact of adoption of ITC on firm market values. The event study shows that ITC announcements result in positive market returns around the date of announcement.	Bose and Leung (2019)
Information technology (IT)	The study tests the maturity and scope of an IT event to examine the suitability of short- versus long-term abnormal returns. The result shows the event value for cases characterized by high maturity and low scope; however, long-term abnormal returns are realized for events involving low maturity and/or high scope.	Barua and Mani (2018)
Internet-of-Things initiatives (IoT)	This study shows that IoT initiative announcements can increase business value, and that this increase is higher when the announcement includes information about specific IoT technologies or when the firm has ongoing IoT projects; this effect is more pronounced for non-IT firms.	Huang et al. (2019)
Steve Jobs	The authors seek to evaluate the value investors placed on Steve Jobs by studying stock market reactions to his death. The results show that the three-day cumulative abnormal returns for Apple are -5.76%, translating into an approximately 20-billion-dollar loss in wealth for investors.	Chatjuthamard et al. (2017)
Information security certification announcement	This is an empirical study of the effects of information security management program investments on shareholder value, showing that the associated abnormal stock market reaction is both positive and statistically significant.	Deane et al. (2019)

Market Reaction by Measuring Abnormal Returns

An abnormal stock return is the difference between the actual return and the expected return and is usually used as an indicator of the impact of an event, unexpected information, or occurrence. In addition, the expected return is estimated by using the average market performance over a given period preceding the event, typically using broad indexes such as S&P 500, NASDAQ, and Dow Jones. When the actual return is higher than the expected return, the abnormal return is positive, meaning that the stock market has a positive reaction to the event, and vice versa.

When deriving the abnormal return, there are two key factors to be aware of: (1) the definition of the event window, and (2) the expected return estimation if the unexpected event does not occur. The event window around the event day, Day 0, is usually noted as (-x,+y), where x is the number of days before the event day and y is the number of days after the event day. Moreover, according to Konchitchki and O'Leary (2011), short windows are recommended, such as (-1,+1) or (-2,+2). This is to prevent a confounding effect, which may cause confusion about the causality between the abnormal return and the event.

To estimate the expected return, we construct a regression model based on the estimation period; the estimated coefficients from this regression are used to calculate the expected value over the event window in which the stock price is adjusted (Subramani & Walden, 2001). Additionally, the length of the estimation period is suggested to be 200 days preceding the event window (Roztocki and Weistroffer, 2011). The above works are all essential as we seek to measure abnormal returns for one event.

Content Analysis

Content analysis is a text-mining methodology which examines and judges the tone and influence of text by analyzing the characteristics of its content. To facilitate more objective analysis, content characteristics are quantified into numbers and percentages for better comparison by using the representative bag-of-words scheme (Harris, 1954), in which every word in each news release is placed in a list after removing duplicate words, after which a matrix is constructed to show the frequency of each word in each news item.

In addition, to judge the tone of the news, positive and negative words are identified according to the categories of the Harvard-IV-4 psychosocial dictionary (Tetlock, 2007). Li (2008) shows that firms with a large increase in negative words in annual reports experience significantly negative returns. Likewise, Tetlock (2007) indicates that negative words have a stronger correlation with the stock returns than positive words. Consequently, by quantifying the text characteristics, the stock price fluctuation can be predicted.

Another characteristic of news is readability. Higher readability makes it easier for the audience to understand and perceive the content. Although the judgment of the readability sometimes depends on the intended audience (Schriver, 1989), vocabularies used in the content still largely determine the readability of the content (Pitler & Nenkova, 2008). Measures of readability include the Flesch-Kincaid grade level, Gunning Fog index, Coleman-Liau index, automated readability index, and SMOG (Ghose & Ipeirotis, 2011). Because these metrics each have their own pros and cons, in this study, we use all of them to evaluate the readability of the coverage of news announcements with respect to AI implementation.

Research Hypotheses

Given the above discussion, we know that artificial intelligence (AI) is a popular information technology. Thus, it is natural to consider the influence on investors' views on business value when companies announce artificial intelligence implementations. In particular, given the popularity of online media, investors are able to update information immediately and easily. Accordingly, we propose the fundamental hypotheses in response to the following research questions:

> Research Question 1 (RQ1): How do AI implementation announcements impact stock prices of the announcing companies?
>
> Research Question 2 (RQ2): What characteristics of AI implementation announcements are related to abnormal returns?

Hypothesis 1

First and foremost, our study is based on signaling theory (Connelly et al., 2011), indicating that the sender determines whether and how to transmit the information and the receiver determines how to interpret it. In the context of this study, the sending party is the news about the AI implementation announcement, and the receiving party is the investors who would interpret the announcement and react based on their final evaluation. The event study methodology can catch the reaction of the receiving party to understand the signal performance of the sending party.

Next, according to the resource-based view (theory), a company gains a competitive advantage only by using heterogeneous and immobile resources; that is, valuable, rare, inimitable, and unsubstitutable resources are competitive strengths that other competitors cannot duplicate (Barney, 1991). As information technologies rapidly advance, it has become more difficult for companies to develop their own competitive advantages. However, artificial intelligence, a technology that has seen considerable advancements of late, can boost the competitive advantages of companies. In general, AI can generate competitive advantages to companies because of its capability to transform data into useful information. Such information is unique to companies and could facilitate their long-term growth. In addition, even though AI resources, such as AI scientists and AI software, are easy to obtain, the well fit between firm capability and AI capability is uncertain. However, the investors, being aware of the announcement, are still expecting that AI could bring benefits for the AI-implemented firms. The investors believe that these firms investing new resources in their organizations can handle business efficiently and create new competitive advantages in their industries.

As a result, announcements of AI implementation constitute positive signals, which causes investors to believe that the company is more competitive than its rivals and is going to pay more for higher future returns. This leads to the first hypothesis of this study:

> Hypothesis 1 (H1): The abnormal return for an announcement of AI implementation is positive.

Hypothesis 2

News coverage includes different situations. For example, companies may or may not reveal detailed information in AI implementation announcements. A variety of reasons may lead a company to decide not to disclose detailed information, including political factors and confidentiality considerations. However, from the perspective of investors, what they receive is only partial information, which can be a concern due to the two critical factors of AI implementation: complexity and considerable resource costs.

The implementation of new information technology, especially in artificial intelligence, represents a significant determination—even a revolution—for a company. If a company's announcement includes detailed information, investors may perceive that the implementation is well-thought-out and also affordable for the company; in addition, the AI implementation plan may be complete. As a result, we present a hypothesis based on the discussion above:

> Hypothesis 2 (H2): The abnormal return for an announcement of AI implementation with detailed information is higher than that without.

Hypothesis 3

Announcements sometimes mention technological partnerships. One possible reason for cooperation with another company is to create synergy and perform better by combining complementary assets. In particular, when a company is willing to ally with an IT company and thus adequately exploit IT assets, its performance can be maximized (Laudon & Laudon, 2011). In addition, based on the argument of transaction cost theory, companies focus on the jobs they do well and leave others to their business partners (Williamson, 1981). They know that the transaction (external) cost of trading with business partners is less than the internal cost. It thus stands to reason that investors would be interested in companies saving costs by establishing technological partnerships with IT professionals.

Since AI implementation is a time- and resource-consuming process, and a complicated and specialized step, when a company announces a technological partnership for AI implementation, seeking assistance from professional companies, investors interpret such action as a positive signal. In other words, the signal from such a technological partnership means that the company not only may perform better in the future but also has a strong determination to implement AI well. This leads to the following hypothesis:

> Hypothesis 3 (H3): The abnormal return for an announcement of AI implementation that mentions a technological partnership is higher than that without any mention of such a partnership.

Hypothesis 4

Following the hypothesis about partnerships with IT companies, we discuss the IT companies themselves. We know that the expectations for IT-intensive industries are greater than those for non-IT-intensive ones (Im et al., 2001), because IT-intensive industries have abundant experience utilizing IT and thus are expected to be able to implement novel information technologies more easily and efficiently.

As a result, IT companies who announce AI implementation are expected to perform better than non-IT companies due to investors' familiarity with information technologies. However, it should be noted that we cannot consequently determine that the abnormal returns of IT companies would be better than those of non-IT companies (Dos Santos et al., 1993). To sum up, we focus on the differences between industry types and propose two opposite hypotheses:

> Hypothesis 4a (H4a): The abnormal returns for announcements of AI implementation in IT companies are higher than those in non-IT companies.

> Hypothesis 4b (H4b): The abnormal returns for announcements of AI implementation in IT companies are lower than those in non-IT companies.

Hypothesis 5

Though companies announced that they would like to implement AI into their workplaces, investors may have an interest in what types of AI they want to conduct. AI possesses various capabilities to solve different problems in companies, leading to different firm performances (Wamba-Taguimdje et al., 2020). As a result, we can generally classify AI capabilities into two types, cost reduction and revenue improvement. Since the two types of AI could lead to two different firm performances, we want to know if different AI capabilities influencing different firm performances would bring different abnormal returns. In other words, if investors read an AI implementation announcement which has revealed what type of AI capabilities the company wishes to implement, they might have a different perception about this AI investment. Therefore, we consider that they should have a different reaction to the company's stock price. Since there are no past studies to discuss this concern, we thereby propose the following hypothesis:

> Hypothesis 5 (H5): The abnormal return for an announcement of AI implementation is different from AI capabilities (cost reduction and revenue improvement).

Hypothesis 6

Next, we are curious about the influence of the content characteristics of the announcements. Tetlock (2007) shows that negative words have a stronger correlation with stock returns than positive words. Hence, we have Hypotheses 5a and 5b for positive and negative words, respectively.

In addition, we believe that there are still other key factors that impact abnormal returns. Referring to Ghose and Ipeirotis (2011), in our investigation we discuss the issue of online reviews and distinguish two content features of announcements—the length and readability of the content—and accordingly propose Hypotheses 5c and 5d.

> Hypothesis 6a (H6a): The frequency of positive words is positively related to the abnormal return for an announcement of AI implementation

> Hypothesis 6b (H6b): The frequency of negative words is negatively related to the abnormal return for an announcement of AI implementation.

> Hypothesis 6c (H6c): The length of the content is positively related to the abnormal return for an announcement of AI implementation.

> Hypothesis 6d (H6d): The readability of the content is positively related to the abnormal return for an announcement of AI implementation.

Research Methodology

In this section, we discuss how we set up the sample and the research processes. We outline the data collection and selection, describe the procedure of the event study methodology, and propose the regression model for content analysis.

Data Collection and Selection

Event Data Collection

First, we defined an event as an announcement of artificial intelligence (AI) implementation in public news media. Moreover, to increase the search efficiency and accuracy, we compiled two groups of keywords related to AI implementation that could be used by the news media (shown in Table 2). Additionally, we downloaded data related to the S&P 500 Index from Compustat, including the components list, tickers, and GISC sectors of the components.

Next, we collected events from Google Search with these keywords and the names of the component companies from January 1, 2014 to February 3, 2019 (about 5 years). We provide three example events in Appendix A.

Table 2. Search Keywords

Action verbs	Conduct, launch, announce, implement, invest
Nouns	Artificial intelligence, machine learning, deep learning, robotics, data analysis

Financial Data Collection

Since our target companies are components in the list of the S&P 500 Index, we took the S&P 500 Index as the benchmark for the overall stock market to estimate the market return. Also, to construct our estimation regression, we took 200 stock trading days as our estimation period, i.e., days -205 to -6 preceding the event day (Day (-205, -6)). Thus, we downloaded the daily close prices and returns of the S&P 500 Index and companies of the events from January 1, 2012 to February 3, 2019 from YAHOO! FINANCE.

Event Data Selection

After collecting the data, we identified suitable events for this study considering the following eight conditions, based on the suggestions of Tetlock et al. (2008) and Konchitchki and O'Leary (2011):

1. The company in the news report is a component in the S&P 500 Index.
2. The company name must be in the title or the first 25 words of the announced content.
3. The content length of the news should not be less than 50 words.
4. Duplicate news for the same or similar announcements is eliminated.
5. To build the return estimation regression model, the company must have continuous historical stock prices within the window of trading days (-205, -6) preceding the event day.
6. To reduce the impact of the confounding effect, the date interval for each event of the company should be longer than 11 days (the longest length of the event window in this study). Confounding events should be checked around these AI implementation announcement dates, and none of them have other major events, such as earnings announcement, merger and acquisition, new product release, change of directors, etc. to avoid other possible explanations to the impact of AI implementation announcements on business value.
7. In cases where the news concerns technological partnerships, it is acceptable if only one of the companies is a component in the S&P 500 Index.
8. No thinly traded stocks exist in samples.

Subject to these conditions, we acquired 109 news reports concerning 30 companies (see Appendix B for more information). In accordance with the search keywords of nouns in Table 2, the number of news appearing on artificial intelligence, machine learning, deep learning, robotics, and data analysis are 105, 11, 4, 8, and 28, respectively. In addition, there are 44 announced news, which have been identified as confounding news according to Condition (6).

Event Classification

To test Hypotheses 2, 3, 4a, 4b, and 5, we classified the events into different groups. The first classification is whether the announcement of AI implementation includes detailed information or not. Detailed information concerns (1) AI technologies, including software or hardware, the company employed; (2) the budget of the AI implementation planned by the company; or (3) the purpose of the company's AI implementation. The second classification is whether the announcement mentions a technological partnership or not, and the third classification uses the

GICS sectors to determine whether the company making the announcement is an IT company. Ultimately, the last classification is whether the announcement mentions the two types of AI, cost reduction, and revenue improvement. The two ways of classification are referred to in the studies of Davenport and Ronanki (2018) and nibusinessinfo.co.uk (2021). If the announcement reveals that the company will implement AI for manufacturing or operation management, such as pattern recognition or detection of face masks, we identify it as the former type. In the past, managers recruit lots of manpower to check for flaws in products or carry out the task of epidemic prevention. However, these labors can be replaced by AI, leading to reduce the labor cost and the opportunity for human error, thus improving production processes. Moreover, we identify AI as the latter type if the company does it for services or trades, such as product recommendation, customer identification (Example 2 in Appendix A), or international trade (Brynjolfsson et al., 2019).

We recruited three experts whose professional fields are in information management and economics. Two experts serve in academic institutions and possess over 10 years' industry-academia cooperation experience for AI implementation. One expert is an economist; he has counseled several enterprises on evaluating the feasibility of AI projects. First, two of them are assigned to read all announcements and then tag them as either cost reduction or revenue improvement. One announcement can be determined as cost reduction or revenue improvement if the two experts make the same decision. The simple measure for inter-rater reliability is to calculate an agreement percentage between raters (Allen, 2018). In this stage, the percent agreement of the two experts is about 80 percent. Second, if they do not achieve an agreement, we then seek the third expert for making a final decision. The reason is to increase the number of classifiable events so that more events can be certainly identified as cost reduction or revenue improvement. The announcements which cannot be distinguished by typeare not used to analyze in the test of Hypothesis 5. As a result, the number of events is 55. The results are shown in Table 3.

Data Collection for Content Characteristics

For content analysis, we developed a program to extract the content characteristics of each event, including the length of the content and the number of positive and negative words. We consulted the Harvard-IV-4 psychosocial dictionary (Tetlock, 2007) to identify the categories of positive and negative words.

After that, we collected the readability scores from READABLE.IO, an online tool, and averaged the scores of five indexes: the Flesch Kincaid Grade Level, the Gunning Fog Score, the Coleman Liau Index, the SMOG Index, and the Automated Readability Index. This yielded the average readability scores for each event for analysis.

Event Study Methodology

In this study, we adopt the market model and the relationship between the stock return and the market return, which can be described as

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}, \qquad (1)$$

where R_{it} is the return of stock i on day t; R_{mt} is the market return on day t, which in this study is the S&P 500 Index; α_i is the intercept term; β_i is the systematic risk of stock i; and ε_{it} is the error term, $\varepsilon_{it} \sim N(0, \sigma^2)$.

From Equation (1), $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated through the ordinary least squares method (OLS) by using the actual parameters (R_{it} and R_{mt}) during the estimation period before the event period. Thus, we can calculate the expected return if the event i of event period E does not occur; the expected return can be expressed as

$$E(\hat{R}_{iE}) = \hat{\alpha}_i + \hat{\beta}_i R_{mE}, \qquad (2)$$

Table 3. Event Classification

Classification	Definition	Number of events
1	Includes detailed information	62
	No detailed information	47
2	Mentions technological partnerships	70
	Does not mention technological partnerships	39
3	IT company	59
	Non-IT company	50
4	Cost reduction	34
	Revenue improvement	21

where $E(\hat{R}_{iE})$ is the expected return of event i of event period E; R_{mE} is the market return of event period E; and $E(\hat{R}_{iE})$ simplifies the prediction errors, which means that the expected value of error equals zero ($E(\varepsilon_{iE}) = 0$).

Consequently, we can compute the abnormal return of event i in event period E (AR_{iE}) by Equation (3). Furthermore, to reduce the influences from confounding effects, we calculate the average abnormal return (*ARE*) by Equation (4).

$$AR_{iE} = R_{iE} - E(\hat{R}_{iE}) = R_{iE} - (\hat{\alpha}_i + \hat{\beta}_i R_{mE}) \quad (3)$$

$$AR_E = \frac{1}{N}\sum_{i=1}^{N} AR_{iE}, \quad (4)$$

where AR_E is the average abnormal return of all events in event period *E*, and *N* is the total number of events.

For the t-tests, the standardized abnormal return (SAR_{iE}) and standardized average abnormal return (SAR_E) are calculated as

$$SAR_{iE} = \frac{AR_{iE}}{\hat{S}_i\sqrt{1 + \frac{1}{T_i} + \frac{(R_{mE} - \bar{R}_{mi})^2}{\sum_{\tau=t_1}^{t_2}(R_{m\tau} - \bar{R}_{mi})^2}}} \quad (5)$$

$$SAR_E = \frac{1}{N}\sum_{i=1}^{N} SAR_{iE}, \quad (6)$$

where $\hat{S}_i$ is the residual standard deviation calculated from the estimation period for event i; T_i is the number of estimation days ($T = t_2 - t_1 + 1, t_2 > t_1$), which in this study equals 200 days; $\bar{R}_{mi}$ is the average market return of event i in the estimation period; $R_{m\tau}$ is the market return of day τ in the estimation period; and R_{mE} is the market return of the day of event period E.

Since we have specified the event windows ([$\mathcal{T}_1$, $\mathcal{T}_2$], $\mathcal{T}_2 > T_1$), we value the cumulative abnormal return (CAR_{iE}) of company *i* on event period E by Equation (7) and calculate the cumulative average abnormal return (CAR_E) on event period E by Equation (8).

$$CAR_{iE} = \sum_{E=\mathcal{T}_1}^{\mathcal{T}_2} AR_{iE} \quad (7)$$

$$CAR_E = \sum_{E=\mathcal{T}_1}^{\mathcal{T}_2} AR_E = \frac{1}{N}\sum_{E=\mathcal{T}_1}^{\mathcal{T}_2} CAR_{iE}, \quad (8)$$

where $\mathcal{T}_1$ and $\mathcal{T}_2$ are the starting and ending days of the event window, respectively.

In essence, event study methodology examines whether the average abnormal return (AR_E) and cumulative average abnormal return (CAR_E) over the event period differ significantly from zero. Therefore, we use two t-statistic methods, the traditional method (TM) and the standardized residual method (SRM), as well as a nonparametric method (NPM), the generalized sign test (GS), to examine the statistical significance.

Equations (9) and (10) are the traditional t-statistic methods for the average abnormal return (AR_E) and cumulative average abnormal return (CAR_E), respectively.

$$t_{TM}^{AR} = \frac{AR_E}{\frac{1}{N}\sqrt{\sum_{i=1}^{N}\hat{S}_i^2}} \quad (9)$$

$$t_{TM}^{CAR} = \frac{\sum_{i=1}^{N}\sum_{E=\mathcal{T}_1}^{\mathcal{T}_2}\left(\frac{AR_{iE}}{\hat{S}_i}\right)}{\sqrt{N(\mathcal{T}_2 - \mathcal{T}_1 + 1)}} \quad (10)$$

where $\hat{S}_i^2$ is the residual variance from the estimation period, and the calculation is $\hat{S}_i^2 = \sum_{t=t_1}^{t_2}(\hat{\varepsilon}_{it} - \bar{\hat{\varepsilon}}_i)^2/(T_i - 1)$.

Next, Equations (11) and (12) are the t-statistics using the standardized residual method for examining standardized average abnormal return (SAR_E) and standardized cumulative average abnormal return ($SCAR_E$), respectively.

$$t_{SRM}^{AR} = \frac{\sum_{i=1}^{N} SAR_{iE}}{\sqrt{\sum_{i=1}^{N}\frac{T_i - 2}{T_i - 4}}} \quad (11)$$

$$t_{SRM}^{SCAR} = \frac{\sum_{i=1}^{N}\sum_{E=\mathcal{T}_1}^{\mathcal{T}_2}\left(\frac{SAR_{iE}}{\sqrt{\mathcal{T}_2 - \mathcal{T}_1 + 1}}\right)}{\sqrt{\sum_{i=1}^{N}\frac{T_i - 2}{T_i - 4}}} \quad (12)$$

Equation (13) shows the formula of the test statistics for the generalized sign test (Cowan, 1992).

$$z_{GS}^{CAR} = \frac{w - N\hat{p}}{[N\hat{p}(1-p)^{1/2}]} \quad (13)$$

where *w* is the number of events (stocks) with positive cumulative abnormal returns during the event period. To reckon the *p*-value, a normal approximation to the binomial distribution with the parameters $\hat{p}$ and N is employed, where $\hat{p}=\frac{1}{N}\sum_{i=1}^{N}\frac{1}{(\mathcal{T}_2-\mathcal{T}_1+1)}\sum_{E=\mathcal{T}_1}^{\mathcal{T}_2}s_{iE}$ and $s_{iE}=\begin{cases}1, \text{ if } AR_{iE}>0\\ 0, \text{ otherwise}\end{cases}$.

Through the above equations, we investigate whether the announcement of AI implementation has a significant influence on the abnormal return.

Content Analysis Methodology

For the content characteristics, we developed a Java program to assist in calculating the number of positive/negative words appearing in the news via comparison with the list of positive/negative words from the Harvard-IV-4 dictionary. The program also calculates the number of words in the news release. READABLE.IO (https://readable.com/) is used to calculate the readability score. In addition, we conduct a stepwise regression analysis to investigate whether the characteristics are correlated with the abnormal return.

The dependent variable is the value of the cumulative abnormal return from event period E of event i (CAR_{iE}). We use the CAR_{iE} over the one-day event window (0, 0), three-day event window (-1, +1), five-day event window (-2, +2), seven-day event window (-3, +3), and eleven-day event window (-5, +5) as the dependent variables.

The independent variables are the content characteristics, including the frequency of positive and negative words, the length of the content, and the readability of the content. In addition, to eliminate differences in magnitude among the values of the characteristics, we adopt the logarithm values for the length of the content and the readability of the content. To examine the impact on abnormal returns in the event period, we propose the following regression model, for which the definition of independent variables is shown in Table 4.

$$\begin{aligned}CAR_{iE} &= \alpha_i+\beta_1 freq(Positive\ Words)_{iE}\\ &+\beta_2 freq(Negative\ Words)_{iE}\\ &+\beta_3\log(Content\ Length)_{iE}\\ &+\beta_4\log(Readability)_{iE}+\varepsilon_{iE}\end{aligned} \tag{14}$$

Data Analysis and Results

This section is organized as follows: the first section introduces the influences of the announcements of AI implementation for all samples; the second section presents the effect of news coverage with detailed information; the third section shows the effect of mentions of technological partnerships; the fourth section reports the effect of different types of industries; the fifth section reports the effect of two types of AI; and the sixth section describes the results of the regression analysis from the perspective of sentiment. We draw an overall conclusion in the last section.

Effect of Announcements of AI Implementation

In this study, 109 sample events of announcements of AI implementation are observed in the event period from event day -5 to event day +5, and two statistical methods, the traditional method (TM) t-test and the standardized residual method (SRM) t-test, as well as a nonparametric method (NPM), the generalized sign test (GS), are employed throughout our analysis to test whether the average abnormal returns (AR_E) and the cumulative average abnormal returns (CAR_E) over the event period differ significantly from zero.

We begin with the average abnormal returns (AR_E): Table 5 indicates that the average abnormal return (AR_{E}) for announcements of AI implementation is positive and significant on Day 0 (AR_{E} =0.260%, traditional method t-test=2.1351**, nonparametric method (NPM) GS=1.9214*).

Table 4. Definition of Content Analysis Variables

Variable	Definition
freq (Positive Words)	The number of positive words divided by the number of words in the news release
freq(Negative Words)	The number of negative words divided by the number of words in the news release
log(Content Length)	The logarithm of the number of words in the news release
log(Readability)	The logarithm of the average score from five specific readability indexes

Table 5. Daily AR_E and t-values in Event Period

Event day	N	AR_E (%)	Standard deviation	Traditional method (TM) t-test	Standardized residual method (SRM) t-test	Nonparametric method (NPM) GS
-5	109	-0.021	0.0110	-0.1745	-0.3629	-0.4143
-4	109	0.002	0.0139	0.0140	-0.4050	0.5147
-3	109	-0.151	0.0110	-1.2402	-0.9269	-0.8754
-2	109	0.147	0.0299	1.2106	0.9383	1.1421
-1	109	-0.105	0.0140	-0.8608	-0.1238	-0.9478
0	109	0.260	0.0116	2.1351**	1.4318	1.9214*
+1	109	0.054	0.0163	0.4453	1.6114	1.8141*
+2	109	-0.070	0.0141	-0.5721	-0.4919	-0.6321
+3	109	-0.161	0.0147	-1.3241	-1.0231	-1.0157
+4	109	0.067	0.0154	0.5545	0.2604	0.3477
+5	109	-0.056	0.0170	-0.4635	-0.9407	-0.4421
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test						

Table 6. CAR_E and t-values of all Samples in Event Windows

Event window	CAR_E(%)	TM t-test	SRM t-test	NPM GS
[-1, +1]	0.2092	1.700*	1.680*	1.770*
[-2, +2]	0.2869	1.517	1.506	1.251
[-3, +3]	-0.0251	0.536	0.538	0.637
[-5, +5]	-0.0335	0.060	0.073	0.249
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test				

The CAR_E of the 109 sample events in the 5-day window are presented in Table 6, according to which the CAR_E of the companies announcing AI implementations are positive and significant in (-1, +1) around the event day (CAR_E=0.2092 percent, TM t-test=1.700*, SRM t-test=1.680*.)

Given these findings, we understand that announcements of AI implementation certainly impact AR_E. Furthermore, from Tables 5 and 6, we see that the influence of such announcements is immediate but lasts only for a short period; it occurs on the event day of AR_E and (-1, +1) for CAR_E. Clearly, the first proposed hypothesis (H1) is statistically supported.

Effect of Detailed Information

In this subsection, we divide all event samples into two groups. Of all samples, 62 events are announcements of AI implementation with detailed information and 47 events are without detailed information. The results are shown in Table 7.

Table 7. CAR_E and t-values for Announcements with and without Detailed Information

Event window	With detailed information (n=62)				Without detailed information (n=47)				Difference in CAR_E
	CAR_E (%)	TM t-test	SRM t-test	NPM GS	CAR_E (%)	TM t-test	SRM t-test	NPM GS	
[0, 0]	0.133	0.845	0.118	0.551	0.426	2.240**	2.045**	1.998**	1.260
[-1, +1]	0.303	1.559	1.546	1.241	0.086	0.803	0.791	0.801	-0.490
[-2, +2]	0.839	2.610**	2.589**	1.914*	-0.441	-0.691	-0.681	-0.741	-1.670*
[-3, +3]	0.571	1.691*	1.681*	1.514	-0.812	-1.128	-1.116	-0.987	-1.650
[-5, +5]	0.481	0.627	0.634	0.741	-0.712	-0.750	-0.744	-0.612	-1.150
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test of significance									

The CAR_E for the announcements with detailed information are positive and significant for two event periods: (-2, +2) (CAR_E =0.839 percent, TM *t*-test=2.610**, SRM *t*-test=2.589**, NPM GS=1.914*) and (-3, +3) (CAR_E=0.571 percent, TM *t*-test=1.691*, SRM *t*-test=1.681*); the CAR_E for the announcements without detailed information are positive and significant in one event period: (0, 0) (CAR_E=0.426 percent, TM *t*-test=2.240**, SRM *t*-test=2.045**, NPM GS=1.998**).

Even though the different results from the two-sample *t*-test between the two groups in the CAR_E are only significant in event window (-2, +2) (*t* =-1.670*) (CAR_E with detailed information is higher than that without), we may still conclude that the second hypothesis (H2) is partially supported.

Effect of Technological Partnerships

In this subsection, we divide all event samples into two groups. Of all samples, 70 events of announcements of AI implementation mention a technological partnership; 39 events do not. The results are shown in Table 8.

The CAR_E for announcements that mention technological partnerships are positive and significant for one event period: (-2, +2) (CAR_E=0.453 percent, TM *t*-test=1.770*, SRM *t*-test=1.757*, NPM GS=1.810*); the CAR_E for the announcements that do not mention technological partnerships are positive and significant in one event period: (0, 0) (CAR_E=0.352 percent, TM *t*-test=1.920*, SRM *t*-test=1.505, NPM GS=1.897*).

Observing each of the event windows, there is no significant difference between the two groups in CAR_E (the former is higher than the latter). Therefore, the third hypothesis (H3) is not supported.

Effect of Information Technology Companies

In this subsection, we divide all event samples into two groups. Of all samples, 59 events are announcements of AI implementation by information technology (IT) companies; 50 events are not. The results are shown in Table 9.

Table 8. CAR_E and t-values with and without Mention of Technological Partnerships

Event window	Technological partnerships mentioned (n=70)				Technological partnerships not mentioned (n=39)				Difference in CAR_E
	CAR_E (%)	TM t-test	SRM t-test	NPM GS	CAR_E (%)	TM t-test	SRM t-test	NPM GS	
[0, 0]	0.208	1.305	0.663	1.014	0.352	1.920*	1.505	1.897*	0.560
[-1, +1]	0.186	1.447	1.437	1.241	0.250	0.909	0.892	0.787	0.140
[-2, +2]	0.453	1.770*	1.757*	1.810*	-0.012	0.161	0.162	0.210	-0.660
[-3, +3]	-0.059	0.311	0.313	0.401	0.035	0.477	0.475	0.381	0.120
[-5, +5]	-0.220	-0.572	-0.554	-0.487	0.301	0.732	0.725	0.826	0.560
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test of significance									

Table 9. CAR_E and t-values for IT and non-IT Companies

Event window	IT companies (n=59)				Non-IT companies (n=50)				Difference in CAR_E
	CAR_E (%)	TM t-test	SRM t-test	NPM GS	CAR_E (%)	TM t-test	SRM t-test	NPM GS	
[0, 0]	0.441	2.570**	2.167**	2.013**	0.046	0.271	-0.240	0.311	-1.770*
[-1, +1]	0.281	1.438	1.425	1.243	0.124	0.952	0.940	0.452	-0.370
[-2, +2]	0.601	1.753*	1.738*	1.503	-0.084	0.332	0.335	0.517	-0.890
[-3, +3]	0.292	0.704	0.699	0.614	-0.399	0.025	0.031	0.014	-0.830
[-5, +5]	0.561	0.664	0.668	0.574	-0.735	-0.751	-0.740	-0.547	-1.300
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test of significance									

The CAR_E for the announcements of IT companies are positive and significant for two event periods: (0, 0) (CAR_E =0.441 percent, TM *t*-test=2.570**, SRM *t*-test=2.167**, NPM GS=2.013**) and (-2, +2) (CAR_E =0.601 percent, TM *t*-test=1.753*, SRM *t*-test=1.738*); the CAR_E s for the announcements of non-IT companies are all non-significant in every event period.

From the two-sample *t*-test, the difference between the two groups in CAR_E is significant in event window (0, 0) (t =-1.770*), showing that CAR_E in IT companies is higher than that in non-IT companies. Hence, hypothesis 4a (H4a) is supported.

Effect of Two Types of AI

In this analysis, we cannot divide all event samples into two groups for testing the two types of AI because not all samples mentioned the type of AI. Therefore, there are 34 events of which types are identified as cost reduction and 21 events are revenue improvement. The results are shown in Table 10.

The CAR_E for the announcements with the type of AI for cost reduction are positive and significant for one event period, (0, 0) (CAR_E =0.203 percent, TM *t*-test=1.773*, SRM *t*-test=2.146**, NPM GS=1.844*). In addition, there is no significant difference between the two groups in CAR_E. Therefore, the fifth hypothesis (H5) is not supported.

Analysis from Perspective of Sentiment

In this subsection, we use stepwise regression analysis to explore the effect on cumulative abnormal returns (CAR_{iE}) from the perspective of sentiment. We specify five window periods of CAR_{iE}, with (0, 0), (-1, +1), (-2, +2), (-3, +3), and (-5, +5) as our dependent variables; and the independent variables include the frequency of positive words, the frequency of negative words, the logarithm of the length of the content, and the logarithm of the average readability score of each announcement. A control variable, industry type (IT and non-IT), is also considered in the regression.

The descriptive statistics of all variables are reported in Table 11, and the results of Pearson correlation coefficients for all variables are presented in Table 12. Note that negative word frequency is the only text characteristic that shows a modest negative correlation with $CAR_{i(0,0)}$ (correlation coefficient=-0.1456); the others are all weakly correlated with CAR_{iE} . Hence, we take only $CAR_{i(0,0)}$ into consideration.

Table 10. CAR_E and t-values for Announcements of Two Different Types of AI

Event window	Cost reduction (n=34)				Revenue improvement (n=21)				Difference in CAR_E
	CAR_E (%)	TM t-test	SRM t-test	NPM GS	CAR_E (%)	TM t-test	SRM t-test	NPM GS	
[0, 0]	0.203	1.773*	2.146**	1.844*	0.104	1.040	1.005	1.147	0.808
[-1, +1]	0.283	1.431	1.521	1.329	0.086	0.813	0.791	0.874	-0.471
[-2, +2]	0.394	1.431	0.911	0.729	0.441	-0.592	-0.594	-0.439	-1.263
[-3, +3]	0.401	1.491	1.301	1.103	-0.773	-1.008	-1.011	-0.941	-1.381
[-5, +5]	0.381	0.533	0.434	0.666	-0.689	-0.621	-0.676	-0.549	-1.240
Notes: ***p<0.01, **p<0.05, *p<0.1; two-tailed test of significance									

Table 11. Descriptive Statistics of Variables

Variable	N	Mean	Std. dev.	Min	Max
$CAR_{i[0,0]}$	109	0.0026	0.0116	-0.0235	0.0439
$CAR_{i[-1,+1]}$	109	0.0007	0.0073	-0.0246	0.0238
$CAR_{i[-2,+2]}$	109	0.0006	0.0083	-0.0251	0.0637
$CAR_{i[-3,+3]}$	109	0.0000	0.0063	-0.0223	0.0415
$CAR_{i[-5,+5]}$	109	0.0000	0.0048	-0.0178	0.0205
freq(Positive Words)	109	1.8812	1.1391	0.0000	3.8700
freq(Negative Words)	109	0.6624	0.6572	0.0000	6.3100
log(Content Length)	109	2.7992	0.3306	1.8865	3.4564
log(Readability)	109	1.1249	0.0871	0.8007	1.2690

Table 12. Correlations of Variables

Variable	$CAR_{i[0,0]}$	$CAR_{i[-1,+1]}$	$CAR_{i[-2,+2]}$	$CAR_{i[-3,+3]}$	$CAR_{i[-5,+5]}$
freq(Positive Words)	-0.0861	-0.0094	0.0031	0.0158	-0.0524
freq(Negative Words)	-0.1456	-0.0584	-0.0144	0.0160	0.0878
log(Content Length)	0.0153	0.0393	0.0582	0.0203	0.0590
log(Readability)	-0.0543	0.0821	-0.0444	-0.0314	0.0056

Table 13. Regression Results of $CAR_{i[0,0]}$ with Text Characteristics

Independent variable	Model			
	1	2	3	4
freq(Positive Words)	-0.8940	-1.3016	-1.4095	-1.1125
	(-0.0009)	(-0.0013)	(-0.0015)	(-0.0013)
freq(Negative Words)		-1.7916*	-1.8541*	-1.8471*
		(-0.0031)	(-0.0033)	(-0.0033)
log(Content Length)			0.6171	0.6402
			(0.0021)	(0.0023)
log(Readability)				-0.1854
				(-0.0029)
Control	Industry	Industry	Industry	Industry
Adjusted R^2	0.0000	0.0184	0.0126	0.0035
Notes: Dependent variable=$CAR_{i[0,0]}$; ***p<0.01, **p<0.05, *p<0.1; numbers in the parentheses are beta coefficients; others are t-values.				

In the event window, there are four regression models which value the effect of text characteristics on CAR_{iE}; we add stepwise independent variables into the regression models. The frequency of positive words is the only independent variable in Model 1; the frequencies of positive words and negative words are chosen in Model 2; the frequency of positive words, the frequency of negative words, and the logarithm of the content length are picked in Model 3; the frequency of positive words, the frequency of negative words, the logarithm of the content length, and the logarithm of the average readability score are selected in Model 4.

Table 13 presents the regression results of the four models for $CAR_{i(0,0)}$. The coefficients of the frequency of negative words are negative and significant in three models: Model 2 (*t*-value=-1.7916*, β= -0.0031), Model 3 (*t*-value=-1.8541*, β=-0.0033), and Model 4 (*t*-value=-1.8471*, β=-0.0033). However, these three models have lower explanatory power (lower adjusted R^2). Therefore, Hypothesis 5b (H5b) is only weakly supported.

Conclusion

Based on event study methodology and content analysis, we examine the influences of announcements of AI implementation on the stock returns of companies. The followings findings are summarized in Table 14.

1. For announcements of AI implementation, the average abnormal return (AR_E) is positive and significant only on the event day (Day 0), and the cumulative average abnormal return (CAR_E) is positive and significant only for period (-1, +1) around the event day.
2. The difference between announcements with detailed information and those without detailed information is significant for event window (-2, +2) around the event day.
3. For all event windows, there is no significant difference between announcements that mention technological partnerships and those that do not.
4. The difference between announcements of IT companies and those of non-IT companies is significant for event window (0, 0) around the event day.

Table 14. Results of all Hypotheses

	Hypothesis	Supported?
H1	The abnormal return for an announcement of AI implementation is positive.	Yes
H2	The abnormal return for an announcement of AI implementation with detailed information is higher than that without.	Yes
H3	The abnormal return for an announcement of AI implementation that mentions a technological partnership is higher than that without any mention of such a partnership.	No
H4a	The abnormal returns for announcements of AI implementation in IT companies are higher than those in non-IT companies.	Yes
H4b	The abnormal returns for announcements of AI implementation in IT companies are lower than those in non-IT companies.	No
H5	The abnormal return for an announcement of AI implementation is different from AI capabilities (cost reduction and revenue improvement).	No
H6a	The frequency of positive words is positively related to the abnormal return for an announcement of AI implementation	No
H6b	The frequency of negative words is negatively related to the abnormal return for an announcement of AI implementation.	Yes
H6c	The length of the content is positively related to the abnormal return for an announcement of AI implementation.	No
H6d	The readability of the content is positively related to the abnormal return for an announcement of AI implementation.	No

5. For 55 event windows, there are no significant differences between the two types of AI, cost reduction and revenue improvement.
6. Finally, according to content analysis, only one independent variable—the frequency of negative words—shows a modest negative correlation with $CAR_{(0,0)}$. The other independent variables are all weakly correlated with all of the event windows. In addition, the coefficients for the frequency of negative words are all negative and significant in $CAR_{(0,0)}$ in Models 2, 3, and 4.

Discussion and Conclusion

Summary of Results

First of all, this study provides initial empirical evidence that AI implementation can reveal the business value of AI from the perspective of shareholders. The result of the AR_E is significantly positive on event day and CAR_E is significantly positive in the event window (-1, +1). The market reacts to AI implementation positively and immediately; however, the reaction lasts for only a short period of time. In other words, investors are optimistic about announcements, but they do not really understand the meaning of AI implementation from a long-term perspective. This confirms the observation of Konchitchki and O'Leary (2011).

Likewise, in the three classifications for grouping analysis, the CAR_E of announcements without detailed information are significantly positive on event day, whereas the CAR_E of announcements with detailed information are significant positive in the event windows (-2, +2) and (-3, +3). This shows that when investors receive information short on details, the reaction is short and immediate. In comparison, when they receive detailed information, they need more time to digest it, leading to a delayed positive reaction and a significant difference from announcements without detailed information.

Furthermore, this study also reveals that the CAR_E of IT companies are much larger than those of non-IT companies. In particular on the event day, the difference in CAR_E is significant. This is likely because IT companies are much more familiar with IT-related technologies, suggesting that they will be comparatively successful with AI implementation. Thus, non-IT companies should provide additional evidence to persuade investors and boost their confidence in AI implementation."

The two types of AI, cost reduction and revenue improvement, do not have significant differences on abnormal returns. However, only the CAR_E of announcements for the type of cost reduction in the event window (0, 0) is significant. This indicates that investors may perceive the positive application from AI is to reduce the cost but cannot identify the differences in the two types of AI.

Last, this study also examines the correlation between content characteristics and cumulative average abnormal returns. The result is that the frequency of negative words shows a modest negative correlation to CAR_E, whereas other characteristics are weakly related to CAR_E. Similar to the results of past studies discussed in Section 2.4, negative words are more influential than other types of words. We suggest that managers should pay attention to the stories about their AI implementation written by the news reporters. The announcement with too many negative words might send a negative signal to investors, which may bring a pessimistic perspective and decrease their confidence in AI implementation.

Implications

These results can serve as references for practice and research. Our results show a significant impact on abnormal returns as firms announce their AI initiatives. They must also pay attention to content management for their announced news. In the academic field, this study constitutes a guide for future research so that scholars not only seek to improve AI techniques for the benefit of engineers but also demonstrate the substantial benefits of AI implementations for managers or shareholders.

Limitations and Future Work

Despite our best efforts to optimize this study, these limitations remain:

1. After the explosion in the popularity of social media, posts and online comments have emerged on social media platforms. These personal opinions can influence investor perceptions, sometimes as much as news coverage. That is, it may not be easy to determine whether the results of abnormal returns are from comments on social media or from public media announcements. Although we have attempted to exclude confounding effects, there is no way to guarantee that other important events did not occur during our investigation time windows. Accordingly, if online comments and posts affect stock prices, future work can focus on investigating abnormal returns from social media.
2. Since the data collection of this study is based on announcements in public media, the announcing companies may only proffer partial information about AI implementation; therefore, we may have missed key information, making it difficult to precisely measure the impact of AI implementation. Thus, future work should carefully check information quality, improving the quality of the sample pool.
3. In this study, we mainly adopt abnormal returns to represent the potential value of AI, increasing the overall value of the companies after AI implementation. However, the values of abnormal returns that we have captured are investor perceptions, and thus represent not the actual values but the expected values of investors. Consequently, the results of this study do not directly present the true benefits of AI implementation. In the future, researchers may investigate the relationship between inside variables and AI implementation. Financial indicators can certainly reflect the direct performance of AI implementation.
4. The results of this study are limited by the size of our dataset. We have attempted to collect all the events in the collection period, but the samples we collect comprise only 109 events, which is less than that of past studies. As time passes and AI technology matures, more available related news may appear to enlarge the sample size for further analysis.
5. The results of this study are based on the events of component companies in the S&P 500; therefore, we cannot necessarily infer that non-S&P 500 companies will have the same results. As we collected the events, we found many examples of AI implementation news from start-up IT companies and small healthcare companies. As these companies are not listed in the components of the S&P 500, we did not include them in our sample pool. Hence, future works can collect more events from more target companies, such as components of the S&P 1500, for a more thorough overall investigation of the business value of AI.

Conclusion

This study investigates the business value of AI implementation for companies. Drawing upon the efficient-market hypothesis (Fama, 1970), signaling theory (Connelly et al., 2011), and past empirical studies (Konchitchki & O'Leary, 2011), we adopt event study methodology to analyze 109 announcements of AI implementation from 2014 to 2019. The results show that these announcements have a significantly positive impact on abnormal returns. However, these impacts only last for a short period. This suggests that investors and shareholders are optimistic about AI implementation but do not realize how AI implementation will benefit the enterprises that undertake it.

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About the Authors

Cheng-Kui (Tony) Huang received the Ph.D. degree in information management from National Central University of Taiwan in 2006. He is a professor in the Department of Business Administration, National Chung Cheng University of Taiwan. His current research interests include data mining/business intelligence/big data, information management, e-commerce, and soft computing. He has published papers in *Decision Support Systems*, *IEEE Transactions on Systems, Man, and Cybernetics − Part B*, *European Journal of Operational Research*, *Information Technology and Management*, *International Journal of Information Management*, *Data & Knowledge Engineering*, *Information Sciences*, *Fuzzy Sets and Systems*, *Journal of Systems and Software*, *Computers & Education*, *Knowledge-Based Systems*, *Applied Soft Computing*, *Telematics and Informatics*, *Applied Intelligence*, *Intelligent Data Analysis*, *Kybernetes*, *International Journal of Innovative Computing*, *Information and Control*, and *Expert Systems with Applications*.

Yu-Tsung Lee was awarded a Master's degree by the Department of Business Administration, National Chung Cheng University of Taiwan. His current research interests include information management, artificial intelligence, and financial management.

Appendix A

Examples of Events

Due to space constraints, we only provide these selections.

Example 1

Title: Wells Fargo sets up artificial intelligence team in tech push
Date: Feb. 11, 2017
Publication: REUTERS

Wells Fargo & Co has created a team to develop artificial intelligence-based technology and appointed a lead for its newly combined payments businesses, as part of an ongoing push to strengthen its digital offerings.

Example 2

Title: IBM, Salesforce join AI forces with Watson, Einstein
Date: Mar. 6, 2017
Publication: USA TODAY

Customer relationship management company Salesforce announced Monday an artificial intelligence partnership with IBM aimed at boosting the range of predictive analytics it can provide clients.

Example 3

Title: IBM Ponies Up $240 Million For Watson Artificial Intelligence Lab at MIT
Date: Sep. 7, 2017
Publication: FORTUNE

IBM said on Thursday it will spend $240 million over the next decade to fund a new artificial intelligence research lab at the Massachusetts Institute of Technology.

Appendix B

GISC sectors of all the companies included in the 109 events are shown in Figure B.1 and companies with more than four events are shown in Figure B.2.

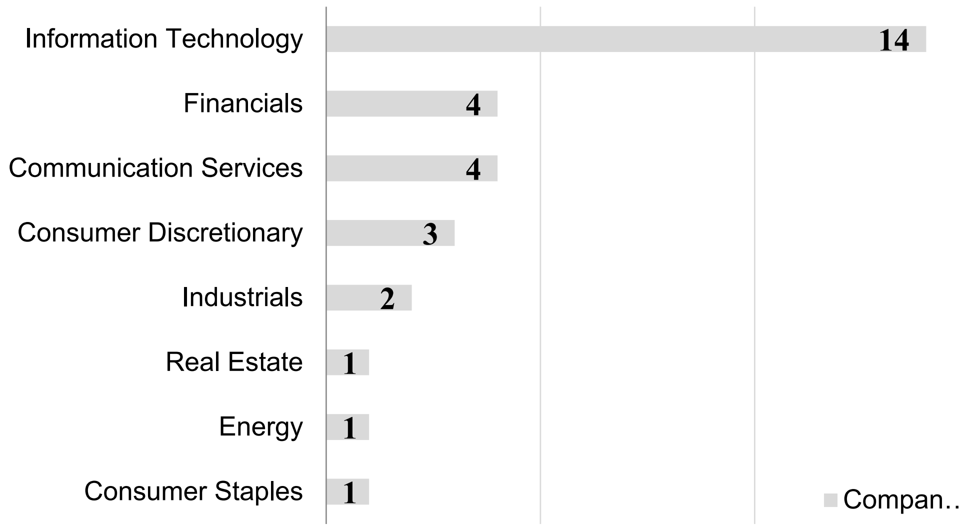

Figure B.1. GISC sectors of all companies included in the 109 events.

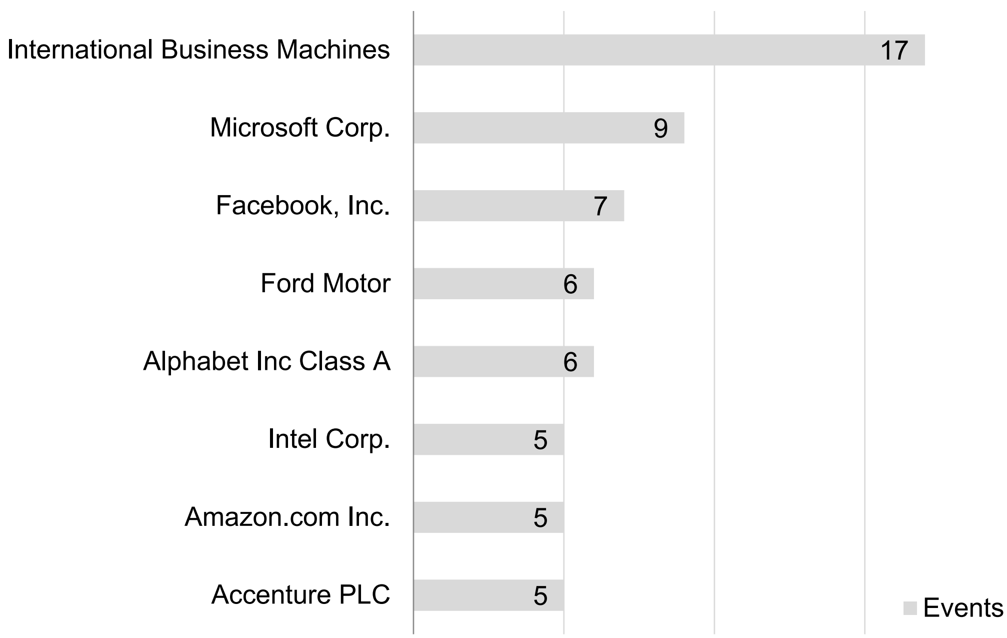

Figure B.2. Companies with more than 4 events.